

Journal Article

You will create a technical article that describes your senior design project. The audience is a technically savvy junior engineer such as yourself who will have to re-implement your project from scratch, or take your design files then improve upon them. The project should introduce the purpose for why it needs to be implemented, then describe the architecture at a high level. After introducing the project and describing the high level architecture with a block diagram, you can go deeper with descriptions of the schematic, board layout, and description of the software. Any measurements of performance or power consumption should be included. Also the estimated cost of the board should be included as well.

Key points to cover:

- Motivation--why did you build it and what does your project do.
- Top level block diagram--be sure to describe architecture and function of sub blocks.
- Schematic--need to relate groups of parts in the schematic to the block diagram.
 - You must discuss the errors that you discovered and how you fixed them (or ignored them).
- Layout--Need to discuss the layout and its relation to schematic.
 - You must discuss errors that you discovered here too and how they were fixed.
 - Did you respin the board? If so, you should show two layouts.
- Software--what did you get working, what did not work?
- Milestones--how far along did you get with the components?
- Reach goals--what is necessary to complete your reach goals or did you reach them?
- Conclusions--this should discuss further work, lessons learned, etc.

Please note that success is not defined by zero errors on the assignment. In my experience, every senior design project had at least some layout errors and sometimes schematic errors. Your job is to communicate your experience and lessons learned.

Please use the IEEE conference proceedings template provided [here](#) when writing your document. It is **required** that you use this format. Note that it is two column and single spaced. When creating sections for your journal article, please recall the sections of a traditional lab report:

- Introduction
- Hypothesis
- Experiment
- Results
- Discussion
- Conclusion
- Works Cited

You may change the titles of your subsections if you feel it communicates your project better. However, you may not leave any information out that is implied by the above subsections. This includes ***Works Cited***.